UNIVERSITE DE BOUMERDES UNIVERSITY OF BOUMERDES	***IGEE*** ***University M'Hamed*** ***Bougara of Boumerdès***	***Last Name:*** ..... ***First Name:*** ..... ***Group:***	***Grade :***

EE321: Computer Architecture
Control Solution (1h)
2023 – 2024

Notes: Answer briefly and clearly using the provided space. No extra sheet will be accepted.

1. Simple CPU Function and Structure (14 pts)

Consider a Little-Endian simple 8-bit CPU with the following specifications: 16-bit address bus, the memory is byte addressable, the instruction list and opcodes are given in Table 1, the instruction formats are given in Table 2. The CPU has 8 data registers (R0 to R7) and 8 address registers (A0 to A7), the register codes are given in Table 3. A0 is the program counter and A7 is the stack pointer. The number of clock cycles for each basic cycle are given in Table 4.

1.1 Give the size in bytes of the instructions: a) `PUSH X`, b) `PUSH Y`, and c) `ADR Ad,As`.

a) PUSH X: 2 bytes.
b) PUSH Y: 3 bytes.
c) ADR Ad,As: 2 bytes. *(1,5 pt.)*

1.2 What is the difference between the two instructions: `PUSH X` and `PUSH Y`. How the CPU makes the difference between the two instructions.

PUSH X stores 1-byte in the stack memory and PUSH Y stores two bytes.
The two instructions have different opcodes. *(1,5 pt.)*

1.3 Using the same format as in Table 1, give the description of the instructions: a) `PUSH X`, and b) `PUSH Y`.

a) PUSH X: $A7 = A7-1, [A7] = X$
b) PUSH Y: $A7 = A7-2, [A7] = Y_{7...0}, [A7+1] = Y_{15...8}$ *(1,5 pt.)*

1.4 Give the HEX content of the registers PC, MAR, and MBR after executing the fetch cycle of the instruction `STD (123H),R0` stored at address 2500H.

The HEX code of the instruction is: 602301 stored at address 2500H. After the fetch cycle we have:
PC = 2503H
MAR = 2502H
MBR = 01H *(1,5 pt.)*

1.5 Give the HEX content of the registers PC, MAR, and MBR after executing the execute cycle of the instruction `STD (123H),R0` stored at address 2500H.

The HEX code of the instruction is: 602301 stored at address 2500H. After the fetch cycle we have:
PC = 2503H
MAR = 0123H
MBR = R0 *(1,5 pt.)*

1.6 Which of the following states: IAC, IF, IOD, OAC, OF, DO, and OS, are involved in the instructions: a) `STD (Y),Rd`, b) `INC Ad`, and c) `MOV Ad,Y`?

a) STD (Y), Rd: IAC, IF, IOD, OAC, and OS.
b) INC Ad: IAC, IF, IOD, and DO.
c) MOV Ad, Y: IAC, IF, IOD. *(1,5 pt.)*

1.7 Give the number of clock cycles it takes to execute (i.e. fetch and execute) the instructions: a) `PUSH X`, b) `PUSH Y`, and c) `STD (Y),Rd`. Justify.

a) PUSH X,: 10 clock cycles, 4 for opcode fetch, 3 for operand fetch, and 3 for data store.
b) PUSH Y: 16 clock cycles, 4 for opcode fetch, 6 for operand fetch, and 6 for data store.
c) STD (Y),Rd: 13 clock cycles, 4 for opcode fetch, 6 for operande fetch, and 3 for data store. *(1,5 pt.)*

1.8 Assuming that we want to add the two instructions `POP X` and `POP Y` to the instruction list given in Table 1, propose an opcode and a description for: a) `POP X`, and b) `POP Y`.

It does not make sense to have those two instructions. *(1,5 pt.)*

1.9 Consider the following memory content starting from 2500H: D0H 23H 01H E0H 45H B0H B1H B2H F0H 10H 0FH FFH. give the assembly program stored at address 2500H.

```
1. 2500H: PUSH 23H      D023
2. 2502:  NOP           01H
3. 2503H: PUSH B045H    E045B0H
4. 2506H: POP R1        B1H
5. 2507H: POP R2        B2H
6. 2508H: HALT          F0H
```
(2 pts.)

2. Stack Memory (6 pts)

2.1 Give one advantage and one disadvantage of the stack memory

Advantage: No need to address manipulation.
Disadvantage: Not random-access memory. *(1,5 pt.)*

2.2 Explain briefly why we can not use the ram to store the return address when executin a call to a subroutine.

If we use a ram, we will have to save the used address, so we didn't solve the problem. *(1,5 pt.)*

2.3 Consider the stack of Z80 processor and SP=3F07H. Assuming register B=B1H, register C=C1H, and the content of the memory starting from address 3F00H is: 00H 01H 02H 03H 04H 05H 06H 07H 08H 09H 0AH, give the content of the memory starting from address 3F00H after executing: a) `POP BC`, and b) `PUSH BC`.

a) Content of the memory starting from address 3F00H: 00H, 01H, 02H, 03H, 04H, 05H, 06H, 07H, 08H, 09H 0AH.
b) Content of the memory starting from address 3F00H: 00H, 01H, 02H, 03H, 04H, C1H, B1H, 07H, 08H, 099H 0AH. *(1,5 pt.)*

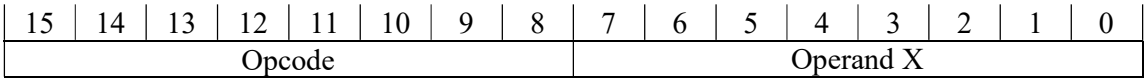
2.4 Consider the stack of Z80 processor and SP=3F07H. Assuming register B=B1H, register C=C1H, and the content of the memory starting from address 3F00H is: 00H 01H 02H 03H 04H 05H 06H 07H 08H 09H 0AH, give the content of the registers SP, MAR and MBR, after executing: a) `POP BC`, and b) `PUSH BC`.

a) SP = 3F09H, MAR=3F08H, and MBR=08H.
b) SP = 3F05H, MAR=3F06H, and MBR=B1H. *(1,5 pt.)*

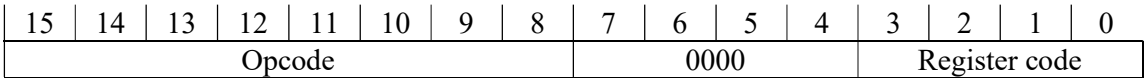
Instruction	Opcode (binary)	Instruction Size	Description
MOV Rd, X	$0001C_3C_2C_1C_0$	2 bytes	Rd = X
MOV Ad, Y	$0001C_3C_2C_1C_0$	3 bytes	Ad = Y
MOV Rd, Rs	$0010C_3C_2C_1C_0$	2 bytes	Rd = Rs
PUSH Rd	$0011C_3C_2C_1C_0$	1 byte	A7 = A7-1, [A7] = Rd
PUSH Ad	$0011C_3C_2C_1C_0$		
PUSH X	11010000		
PUSH Y	11100000		
LDD Rd, (Y)	$0100C_3C_2C_1C_0$		Rd = [Y]
LDR Rd, (As)	$0101C_3C_2C_1C_0$		Rd = [As]
LDR Ad, (As)	$0101C_3C_2C_1C_0$		Ad = [As]
STD (Y), Rd	$0110C_3C_2C_1C_0$		[Y] = Rd
STD (Ad), Rs	$0111C_3C_2C_1C_0$		[Ad] = Rs
ADI Rd, X	$1000C_3C_2C_1C_0$		Rd = Rd + X
ADI Ad, Y	$1000C_3C_2C_1C_0$		Ad = Ad + Y
ADD Rd, (As)	$1001C_3C_2C_1C_0$		Rd = Rd + [As]
ADR Rd, Rs ADR Ad, As	$1010C_3C_2C_1C_0$		Rd = Rd + Rs Ad = Ad + As
POP Rd POP Ad	$1011C_3C_2C_1C_0$		
INC Rd INC Ad	$1100C_3C_2C_1C_0$		Rd = Rd + 1 Ad = Ad + 1
HLT	11110000		Stop the execution of the program

Rd =Data register (it can be any data register).
Rs = Data register (it can be any data register).
Ad = Address register (it can be any address register).
As = Address register (it can be any address register).
$C_3C_2C_1C_0$ = Binary code of the operand (Rd / Ad) (see Table 3).
X =8-bit 2's complement value.
Y =16-bit 2's complement value.
[…] = Content of memory location.

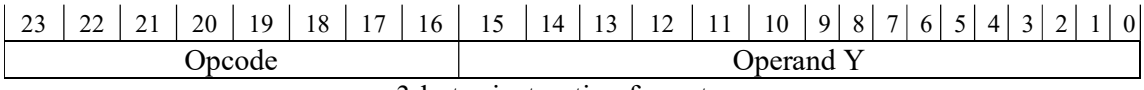
Table 1: Instructions description.



2-bytes instruction format.



2-bytes instruction format.



3-bytes instruction format

Table 2: 2-bytes and 3-bytes instructions formats.

Register	R0	R1	R2	R3	R4	R5	R6	R7
Code	0000	0001	0010	0011	0100	0101	0110	0111

Register	A0	A1	A2	A3	A4	A5	A6	A7
Code	1000	1001	1010	1011	1100	1101	1110	1111

Table 3: Binary codes of data and address registers.

Cycle	Number of clock cycles
Opcode fetch	4 clock cycles
1-byte operand fetch	3 clock cycles
Memory read	3 clock cycles
Memory write	3 clock cycles
Do operation	1 clock cycle

Table 4: Number of clock cycle for each basic cycle.